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Expanding Valley Fever Forecasting in California with Environmental LSTMs

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Introduction

Background:
Valley Fever (*coccidioidomycosis*) is an infectious respiratory disease endemic to the southwestern United States. It spreads when soil containing the fungus *Coccidioides* is disturbed and its spores are inhaled.

Each year, approximately more than 20,000 Valley Fever cases are reported in the U.S., with the number only growing. While symptoms are usually only mild/flu-like, a small portion of cases are severe, with some even resulting in death.

Environmental factors such as precipitation, temperature, wind, fire, and pesticide-use may influence when and where case rates rise. Through predictions we may anticipate outbreaks and case surges before they occur, allowing us to employ effective action.

Research Objectives:

- Identify which environmental factors are the strongest drivers of Valley Fever incidence.
- Effectively and accurately forecast Valley Fever across eight counties in California.

Data

We focus on eight counties in California: Fresno, Kern, Los Angeles, Orange, San Diego, San Luis Obispo, Tulare and Ventura.

For each county we have monthly data* from 2001-2024 on the following environmental variables:

- Precipitation
- Humidity
- Wind Speed
- Wind Run
- Wind Event Count
- Soil Temperature
- Air Quality
- Wildfires
- Fungicide Usage
- Rodenticide Usage

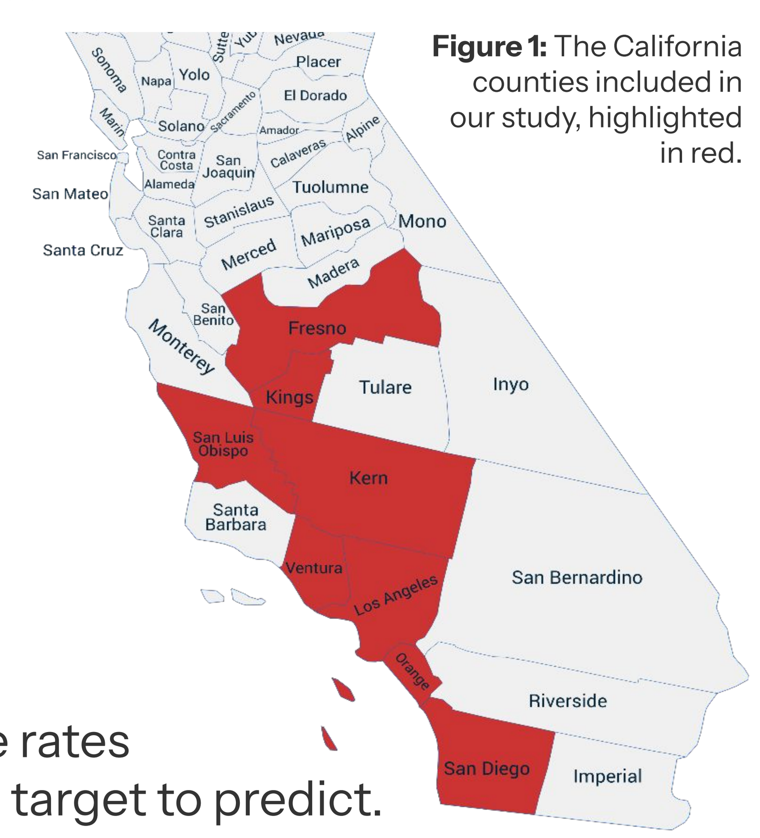

Figure 1: The California counties included in our study, highlighted in red.

We use the Valley Fever case rates per 100,000 residents as our target to predict.

*Data provided by the California Department of Public Health (CDPH), California Irrigation Management Information System (CIMIS), CAL FIRE, California Pesticide Information Portal (CalPIP) and the U.S. Environmental Protection Agency (EPA).

Methodology

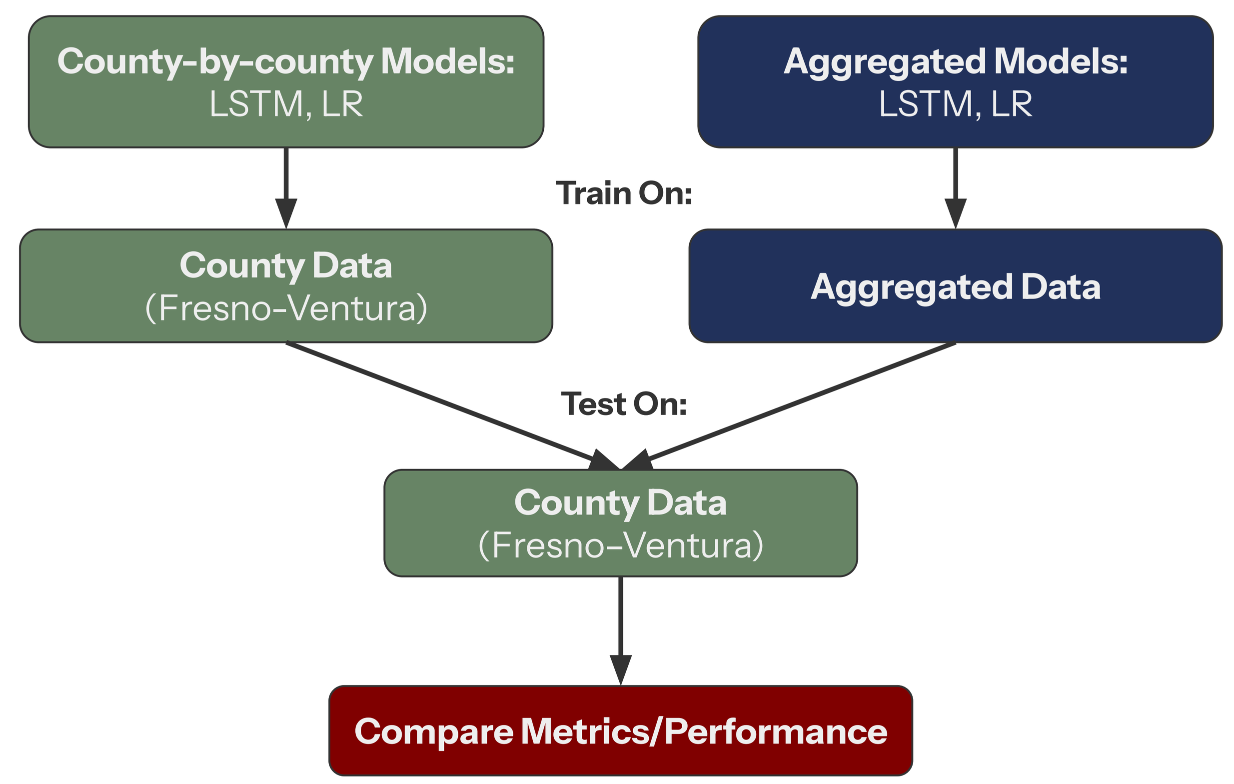

Figure 2: Modeling pipeline comparing county-by-county and aggregated LSTM and linear regression models.

For our research, we evaluated **long short-term memory (LSTM)** models and compared them with **linear regression (LR)** baselines. For both model types, we investigated two forecasting approaches:

County-by-county: Each model was trained and tested using data from the same county.

Aggregated: Each model was trained on a monthly dataset created by aggregating the mean or median feature values across all eight counties, then evaluated separately on each county.

County	First Highest	Second Highest	Third Highest
Fresno	Humidity	Soil Temperature	Wind Event Count
Tulare	Soil Temperature	Fungicide Usage	AQI PM25
Kern	Soil Temperature	AQI PM10	Rodenticide Usage
Ventura	Fungicide Usage	AQI PM10	Rodenticide Usage
Orange	AQI PM25	Precipitation	AQI PM10
San Luis Obispo	Wildfire Acres Burned	Wind Event Count	Fungicide Usage
San Diego	Precipitation	Humidity	Rodenticide Usage
Los Angeles	AQI PM10	Wind Event Count	Humidity

Table 1: Three most influential environmental predictors for each county, ranked by permutation feature importance.

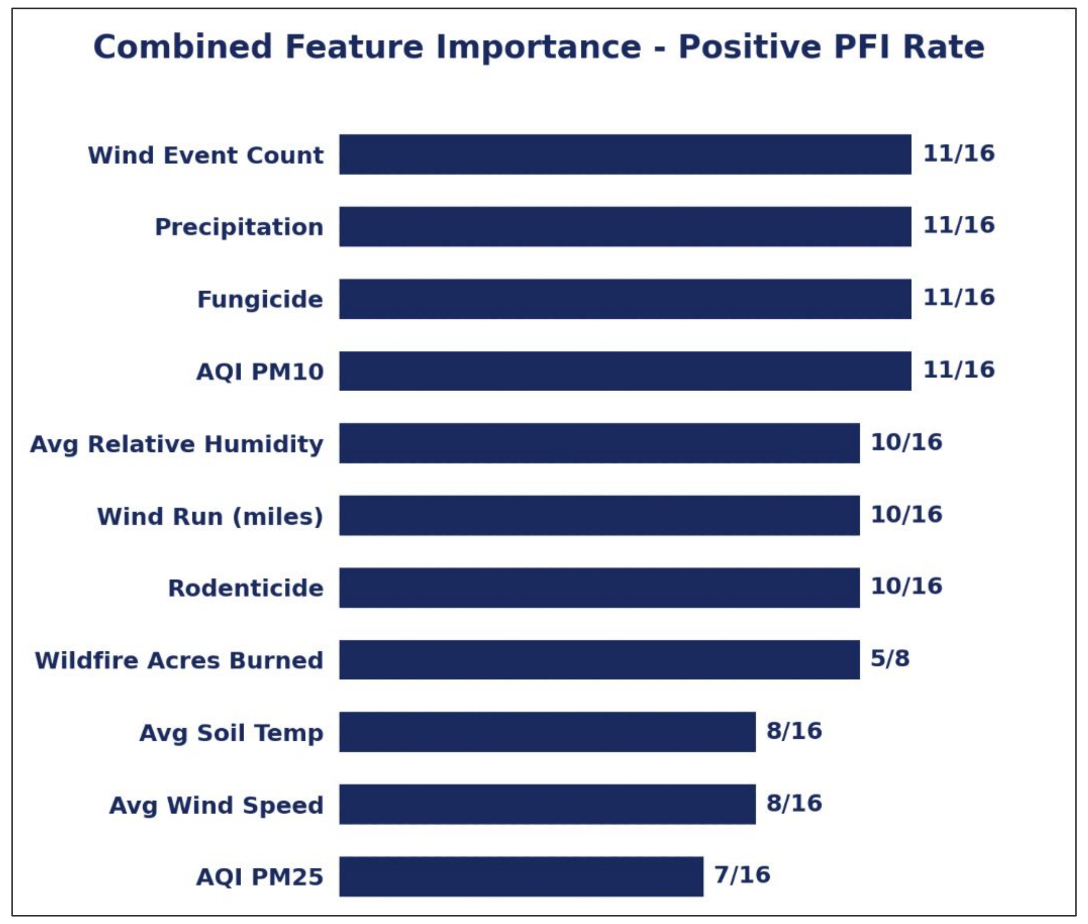

Figure 3: Frequency of positive permutation feature importance across county models (16 evaluations total; wildfire assessed in 8).

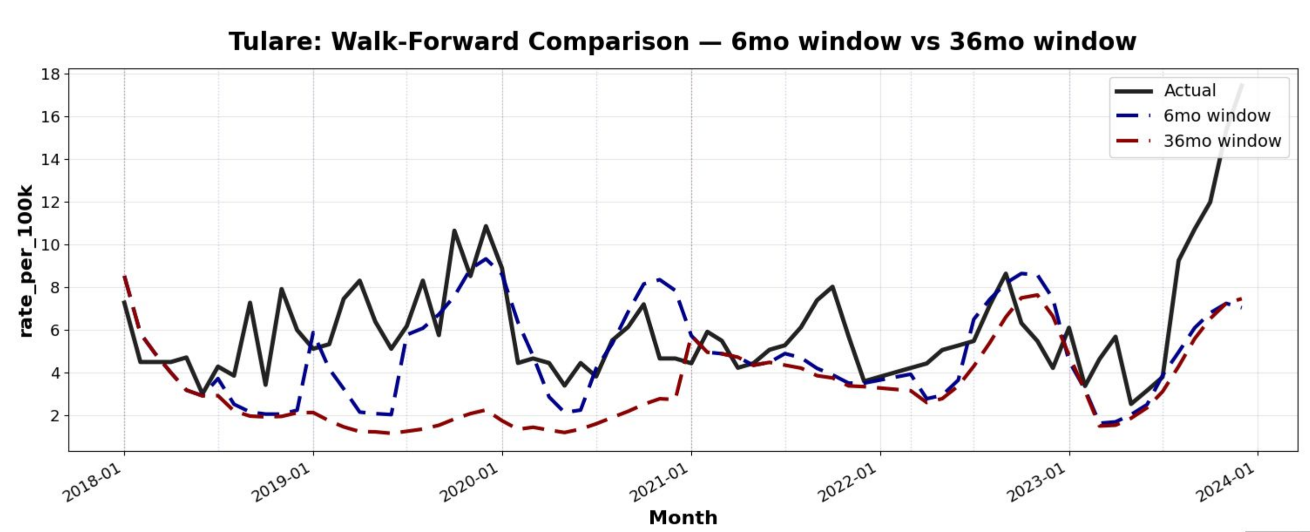

Figure 4: Observed and predicted Valley Fever rates for Tulare county using walk-forward LSTM models with 6-month and 36-month training windows.

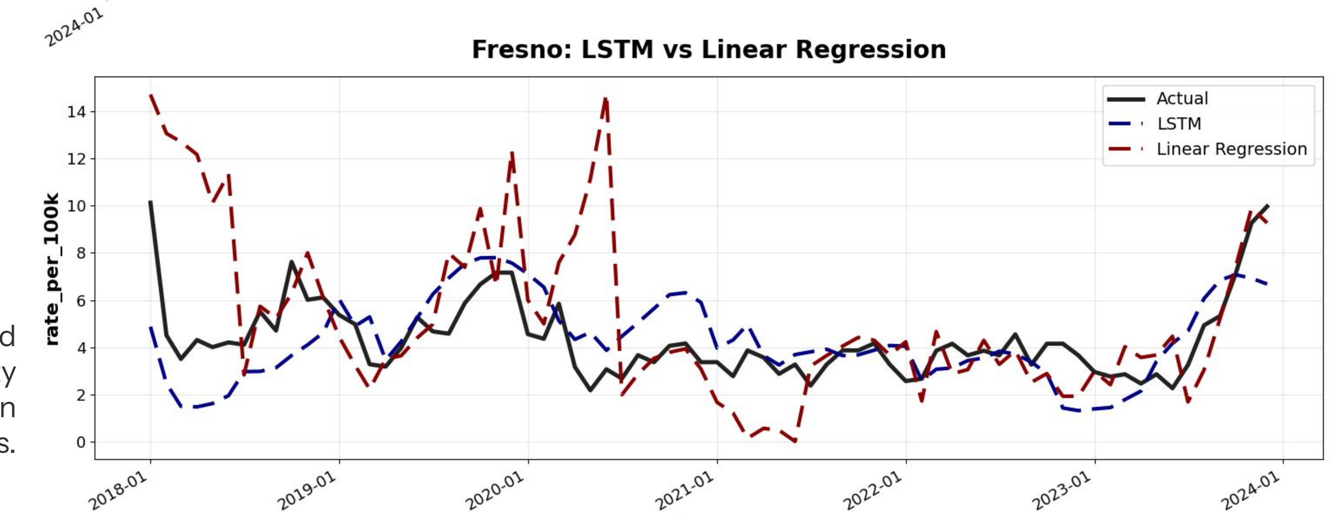

Figure 5: Observed and predicted Valley Fever rates for Fresno county using LSTM and linear regression models.

Results

County	Linear Regression RMSE	Aggregate LSTM RMSE	County-by-county LSTM RMSE
Fresno	2.737	4.276	**1.607**
Tulare	5.573	3.995	**3.098**
Kern	20.168	24.724	**11.201**
Ventura	**1.853**	3.714	1.967
Orange	**0.234**	2.914	0.276
San Luis Obispo	7.141	7.316	**4.074**
San Diego	**0.384**	2.137	0.558
Los Angeles	**0.214**	2.626	0.309

Table 2: Average RMSE comparison across all counties for LR, aggregated LSTM, and county-by-county LSTM models over six-month test periods from 2016–2024.

Conclusions

The importance of features varied by county, supporting the use of county-specific feature selection. Wind events, precipitation, fungicide use and AQI PM10 were among the most consistently influential features. Although LR models achieved competitive RMSEs, the county-by-county LSTMs demonstrated much stronger qualitative performance, better capturing seasonality, temporal variations and dips/peaks in the Valley Fever rates. The county-by-county LSTMs also generally outperformed the aggregated LSTMs, suggesting that county-specific training better conserves local patterns. Shorter training windows were more responsive to recent changes, while longer windows produced smoother forecasts.

Future Work

- Train the aggregate models using data from **all** California counties rather than only eight.
- Group counties into three geographic regions: inland, lower coast, and upper coast.
- Apply feature-specific lags to capture delayed effects.
- Impute or find missing fire data.

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References